Student Presentation Topics - Foundational and Recent Papers

Paper Selection Instructions

1. Choose ONE paper from the list below
2. By the announced selection deadline: Confirm your choice with the instructor
3. By the announced slide deadline: Submit your presentation slides
4. Final session: Present during the two-hour presentation session

Presentation Requirements

- Duration: 8 minutes (strict), followed by 3 minutes of questions
- Format: Slides (PDF or PowerPoint)
- Content: See *Presentation Guidelines*

Paper 1: Brownian Motion and the Atomic Theory (1905)

Author: Albert Einstein Title: “On the motion of small particles suspended in stationary liquids required by the molecular-kinetic theory of heat” Journal: Annalen der Physik Year: 1905

Why This Paper?

- Foundational paper providing evidence for atoms
- Explained Brownian motion as molecular collisions
- Quantified diffusion at molecular scale
- Direct experimental validation by Jean Perrin (1908)

Key Topics to Cover

- Historical context: Debate over atomic theory
- Einstein’s theoretical prediction: $D = kT/(6\pi\eta r)$
- Relationship between diffusion, temperature, particle size
- Perrin’s experimental validation
- Impact on soft matter and colloidal science
- Stochastic processes in physics

Connection to Course

- Lecture 1: Thermal energy scales (kBT)
- Foundational for understanding mesoscopic systems
- Entropy and fluctuations in soft matter

References

- Einstein, A. Ann. Phys. 17, 549 (1905)
- Perrin, J. Ann. Chim. Phys. 18, 5 (1909)
- Wikipedia: Brownian motion

Paper 2: Entropic Forces and Liquid Crystal Ordering (1949)

Author: Lars Onsager Title: “The effects of shape on the interaction of colloidal particles” Journal: Annals of the New York Academy of Sciences Year: 1949

Why This Paper?

- Counter-intuitive: More order increases entropy!
- Shape alone can cause phase transitions
- Foundation for liquid crystal physics
- Shows power of entropy in soft matter

Key Topics to Cover

- Isotropic vs nematic phases
- Hard-rod model (purely repulsive interactions)
- Entropic ordering mechanism
- Role of packing fraction
- Translational vs orientational entropy
- Modern applications in colloidal liquid crystals

Connection to Course

- Lecture 1: Entropy-driven behavior
- Mesoscopic structure determining properties
- Phase transitions without attractive forces

References

- Onsager, L. Ann. N.Y. Acad. Sci. 51, 627 (1949)
- Frenkel, D. Physica A 263, 26 (1999)
- Recent work on DNA liquid crystals

Paper 3: Polymer Dynamics - Reptation (1971)

Author: Pierre-Gilles de Gennes Title: “Reptation of a Polymer Chain in the Presence of Fixed Obstacles” Journal: Journal of Chemical Physics Year: 1971

Why This Paper?

- Revolutionary model for polymer rheology
- Nobel Prize work (1991)
- Molecular picture for viscoelasticity
- Explains dramatic viscosity changes

Key Topics to Cover

- Reptation (snake-like motion) concept
- Tube model for entangled polymers
- Scaling laws: $D \sim N^{-2}$, $\tau \sim N^3$
- Entanglement length and molecular weight
- Experimental validation
- Impact on polymer processing

Connection to Course

- Lecture 5: Viscoelastic behavior
- Lecture 6: Rheology and relaxation times
- Microscopic origin of macroscopic properties

References

- de Gennes, P.G. J. Chem. Phys. 55, 572 (1971)
- Doi, M. & Edwards, S.F. "The Theory of Polymer Dynamics" (1986)
- Nobel lecture (1991)

Paper 4: Auxetic Materials (1987)

Author: Roderic S. Lakes Title: "Foam Structures with a Negative Poisson's Ratio" Journal: Science Year: 1987

Why This Paper?

- Counter-intuitive behavior: Expands when stretched!
- Opened field of mechanical metamaterials
- Design-driven material properties
- Engineering applications

Key Topics to Cover

- Poisson's ratio (normal: positive, auxetic: negative)
- Re-entrant foam structure design
- Mechanical testing results (ν = -0.6)
- Enhanced energy absorption
- Stiffness improvements
- Modern metamaterials inspired by this work

Connection to Course

- Soft matter engineering
- Structure-property relationships
- Mesoscopic design principles

References

- Lakes, R.S. Science 235, 1038 (1987)
- Modern reviews on auxetic materials
- Applications in protective equipment, medical devices

Paper 5: Colloidal Glass Transition (1986)

Authors: P.N. Pusey & W. van Megen Title: "Phase Behavior of Concentrated Suspensions of Nearly Hard Colloidal Spheres" Journal: Nature Year: 1986

Why This Paper?

- Established colloids as model system for glasses
- Bridged soft matter and condensed matter
- Demonstrated glassy dynamics experimentally
- Accessible system for theoretical validation

Key Topics to Cover

- Hard-sphere colloids
- Volume fraction dependence
- Dynamic light scattering technique
- Structural arrest at $\varphi \sim 0.58$
- Glass vs crystal transitions
- Modern applications (real-space imaging)

Connection to Course

- Soft matter phase transitions
- Role of packing and jamming
- Non-equilibrium phenomena

References

- Pusey, P.N. & van Megen, W. Nature 320, 340 (1986)
- van Megen, W. & Underwood, S.M. Nature 362, 616 (1993)
- Review: Hunter & Weeks Rep. Prog. Phys. (2012)

Paper 6: Jamming Concept (1998)

Authors: Andrea J. Liu & Sidney R. Nagel Title: "Jamming is Not Just Cool Any More" Journal: Nature Year: 1998

Why This Paper?

- Unified disparate systems
- Jamming phase diagram concept
- Interdisciplinary impact
- Active research field today

Key Topics to Cover

- Jamming phase diagram (density, load, temperature)
- Granular media, foams, glasses unified
- "Fragile matter" concept
- Unjammed vs jammed states
- Critical packing fraction
- Modern jamming research

Connection to Course

- Lecture 6: G' and G" behavior near jamming
- Transition from fluid to solid
- Non-equilibrium solidification

References

- Liu, A.J. & Nagel, S.R. Nature 396, 21 (1998)
- van Hecke, M. J. Phys. Condens. Matter 22, 033101 (2010)
- Modern jamming reviews

Paper 7: Active Matter - Vicsek Model (1995)

Authors: Tamás Vicsek et al. Title: "Novel Type of Phase Transition in a System of Self-Driven Particles" Journal: Physical Review Letters Year: 1995

Why This Paper?

- Foundational for active matter field
- Simple model with rich behavior
- Non-equilibrium phase transition
- Applications from biology to robotics

Key Topics to Cover

- Vicsek model rules (alignment + noise)
- Phase transition to collective motion
- Order parameter (average velocity)
- Critical noise and density
- Biological examples (flocking, schooling)
- Modern active matter research

Connection to Course

- Beyond thermal equilibrium
- Collective behavior in soft matter
- Pattern formation
- Emergent phenomena

References

- Vicsek, T. et al. Phys. Rev. Lett. 75, 1226 (1995)
- Modern reviews on active matter
- Applications in self-propelled colloids, bacteria

Paper 8: Mechanobiology - Matrix Elasticity (2006)

Authors: Adam J. Engler, Shamik Sen, Dennis E. Discher Title: "Matrix Elasticity Directs Stem Cell Lineage Specification" Journal: Cell Year: 2006

Why This Paper?

- Groundbreaking link between mechanics and biology
- Tissue engineering implications
- Showed cells sense and respond to stiffness
- Developmental biology insights

Key Topics to Cover

- Mesenchymal stem cells
- Substrate stiffness ranges:
  - Soft (0.1-1 kPa) $\rightarrow$ neurons
  - Medium (10 kPa) $\rightarrow$ muscle
  - Stiff (30-40 kPa) $\rightarrow$ bone
- Mechanotransduction mechanism

- Myosin II inhibition experiments
- Impact on tissue engineering

Connection to Course

- Lecture 6: Modulus as meaningful parameter
- Biological relevance of "softness"
- Real-world application of rheology

References

- Engler, A.J. et al. Cell 126, 677 (2006)
- Discher, D.E. et al. Science 310, 1139 (2005)
- Modern mechanobiology reviews

Paper 9: Soft Robotics - Multigait Robot (2011)

Authors: Robert F. Shepherd et al. (Whitesides group) Title: "Multigait soft robot" Journal: Proceedings of the National Academy of Sciences Year: 2011

Why This Paper?

- Seminal soft robotics demonstration
- Pneumatic actuation principles
- Novel locomotion strategies
- Inspired new research field

Key Topics to Cover

- Silicone elastomer construction
- Pneumatic network design
- Multiple gaits (undulating, walking)
- Advantages over rigid robots:
  - Impact resistance
  - Squeezing through gaps
  - Safe human interaction
- Modern soft robotics applications

Connection to Course

- Soft matter engineering
- Rubber elasticity in action
- Practical applications of material properties

References

- Shepherd, R.F. et al. Proc. Natl. Acad. Sci. 108, 20400 (2011)
- Reviews on soft robotics
- Medical and search-rescue applications

Paper 10: Shear Thickening - Dynamic Jamming (2012)

Authors: Scott R. Waitukaitis & Heinrich M. Jaeger Title: "Impact-Activated Solidification of Dense Suspensions via Dynamic Jamming Fronts" Journal: Nature Year: 2012

Why This Paper?

- Explained famous "oobleck" behavior
- Novel mechanism (jamming fronts)
- High-speed imaging insights
- Impact-resistant material applications

Key Topics to Cover

- Cornstarch-water suspension properties
- Discontinuous shear thickening
- Dynamic jamming front propagation
- High-speed imaging and force measurements
- Grain jamming mechanism
- Applications in protective materials

Connection to Course

- Lecture 6: Non-Newtonian rheology
- Dramatic material behavior changes
- Stress-dependent properties
- Connection to Paper 6 (jamming)

References

- Waitukaitis, S.R. & Jaeger, H.M. Nature 487, 205 (2012)
- Morris, J.F. Annu. Rev. Fluid Mech. 52, 121 (2020)
- Shear-thickening fluid applications

Paper Selection Process

Step 1: Review All Papers

- Read abstracts and summaries above
- Consider your interests
- Think about which connects to your background

Step 2: Select a Paper

- Email instructor with:
  - First choice
  - Second choice (backup)
- First-come, first-served for popular papers
- Instructor will confirm assignment

Step 3: Deep Dive

- Read paper thoroughly (multiple times)
- Research background/context

- Find supplementary materials
- Identify key figures to present
- Prepare questions you might receive

Step 4: Create the Presentation

- Follow *Presentation Guidelines*
- Include: context, methods, results, significance
- Practice timing (8 minutes strict)
- Submit slides 24 hours before class

Resources

Finding Papers

- University library access
- Google Scholar
- Ask instructor for PDF if access issues

Understanding Papers

- Read related reviews
- Check citation context
- Office hours for clarification
- Discuss with classmates (respectfully)

Creating Presentations

- PowerPoint / Keynote / Google Slides
- Include clear figures from paper
- Explain technical terms
- Visual aids important

Grading Criteria

See *Lectures 7-8* for the full rubric.

The final presentation contributes 60% of the course grade. The split below is within the presentation mark.

Presentation (70%): - Content understanding: 30% - Clarity of explanation: 20% - Visual aids: 10% - Time management: 10%

Discussion (30%): - Questions asked to peers: 15% - Thoughtful responses to questions: 15%

Questions? Contact Dr. Sanjay during office hours or via email.

Paper selection deadline: Announced for the current course run Slides submission deadline: Announced for the current course run Presentations: Final two-hour course session

Course Home: *Course Overview*

Metadata: Author:: Vatsal Sanjay Date published:: Jul 19, 2026 Date modified:: Jul 19, 2026